

Deep Color-Corrected Multiscale Retinex Network for Underwater Image Enhancement

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Abstract—The acquisition of high-quality underwater images is of great importance to ocean exploration activities. However, images captured in the underwater environment often suffer from degradation due to complex imaging conditions, leading to various issues, such as color cast, low contrast, and low visibility. Although many traditional methods have been used to address these issues, they usually lack robustness in diverse underwater scenes. On the other hand, deep learning techniques struggle to generalize to unseen images, due to the challenge of learning the complicated degradation process. Inspired by the success achieved by the Retinex-based methods, we decompose the underwater image enhancement (UIE) task into two consecutive procedures, including color correction and visibility enhancement, and introduce a novel deep color-corrected multiscale retinex network (CCMSR-Net) (code and models are available at <https://indtlab.github.io/projects/CCMSRNet>). With regard to the two procedures, this network comprises a color correction subnetwork (CC-Net) and an MSR subnetwork (MSR-Net), which are built on top of the hybrid convolution–axial attention block (HCAAB) that we design. Thanks to this block, the CCMSR-Net is able to efficiently capture local characteristics and the global context. Experimental results show that the CCMSR-Net outperforms, or at least performs comparably to, 11 baselines across five test sets. We believe that these promising results are due to the effective combination of color correction methods and the MSR model, achieved by jointly exploiting convolutional neural networks (CNNs) and transformers.

Index Terms—Color correction, deep learning, Retinex model, underwater image enhancement (UIE), underwater image processing.

I. INTRODUCTION

UNDERWATER image processing [1], [2], [3] is key to various ocean exploration activities, such as marine biology [4], underwater inspection [5], and navigation of autonomous underwater vehicles (AUVs) [6]. However, images captured in the underwater environment often suffer from degradation problems, attributed to wavelength-dependent attenuation and forward and backward

scattering caused by the water body [7]. Consequently, underwater images frequently exhibit apparent color cast, low contrast, and blurring, which can interfere with downstream tasks, such as object detection [8] and segmentation [9]. Therefore, underwater image enhancement (UIE) plays an important role in addressing that challenge.

Over the years, traditional approaches have been developed, to enhance the visual quality of underwater images, including restoration-based [10], [11], [12] and enhancement-based methods [13], [14], [15], [16]. Restoration-based methods normally aim to estimate the parameters associated with the degradation process. These parameters are then used to recover a clear underwater image. On the other hand, enhancement-based methods primarily focus on adjusting the pixel values contained in an underwater image for purposes of improving the visibility, colorfulness, and contrast of the image. Among these methods, in particular, the Retinex-based approaches [16], [17] have produced impressive results.

The Retinex model, which was originally proposed by Land [18], simulates the color perception of the human visual system (HVS). Many UIE methods [19], [20], [21] were introduced on top of this model. According to the Retinex model, a degraded image can be perceived as the combination of the reflectance and the illumination. The reflectance represents the intrinsic property of a scene, while the illumination is the primary factor which results in image distortion [18]. Given that the effect of the illumination is estimated and removed, the desired reflectance can then be derived. An enhanced image with increased visibility and perceptual quality can be further obtained using the reflectance data.

However, the Retinex model [18], [20] cannot properly handle color cast alone (refer to “MSR” [20] in Fig. 1). Therefore, this model was often brought together with a color correction algorithm [19], [21]. Despite the effectiveness of this strategy being demonstrated, two problems still remain with these methods. First, the predefined procedures of the existing color correction algorithms struggle to adapt to unpredictable scenes. Second, it is challenging to design an appropriate *prior* for illumination estimation while the fixed prior falls short in accommodating diverse degraded phenomena. Hence, those methods usually encountered the challenge in adapting to the diverse characteristics manifested in underwater images, which might lead to under-enhancement or over-enhancement [22].

With the rapid development of deep learning techniques in the field of computer vision [23], [24], [25], they have been applying to the UIE task. Supervised learning approaches

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[26], [27], [28], [29] were used to train a convolutional neural network (CNN) which aimed to map a degraded underwater image to its clear counterpart. The performance of these approaches severely relied on the availability of the annotated training data. However, it was hard to derive the ground-truth (clear) image for a degraded underwater image. Consequently, the limited training data failed to comprehensively capture the intricacies of the degradation process. Since the supervised learning approaches directly learned the nonlinear mapping, they tended to solely focus on the limited training data. As a result, the model trained could not be popularized to unseen images (refer to “UColor” [28] in Fig. 1).

Many efforts were also made on performing the UIE task using semisupervised learning methods [1], [30], [31], [32], [33] and unsupervised learning approaches [34], [35]. Although these methods avoided the requirement of the annotated underwater images, they were difficult to train and usually produced dissatisfying results (refer to “Zero-Shot” [34] in Fig. 1). In particular, the majority of the existing deep UIE methods were designed on top of CNNs and rarely paid attention to modeling the global context. The limited size of effective receptive fields restricted their capability to capture long-range dependencies [36], which are important to image enhancement [37], [38].

Motivated by the success achieved by the Retinex-based methods, we decompose the intricate UIE operation into two consecutive procedures, i.e., color correction and visibility enhancement, rather than directly learning the nonlinear mapping from degraded images to their clear counterparts, performed by supervised learning methods. We propose a novel deep color-corrected multiscale retinex network (CCMSR-Net). This network comprises a color correction subnetwork (CC-Net) and an MSR subnetwork (MSR-Net), in terms of the two procedures. Compared with the traditional color correction algorithms [14], [15], [17], [40], which severely depend on predefined procedures and the meticulous parameter tuning, the CC-Net can be end-to-end trained and therefore is more robust to diverse degraded images.

For purposes of enhancing the visibility and visual quality of degraded images, the MSR-Net is introduced on top of the classical MSR model [20], to estimate and eliminate the illumination. Unlike some traditional Retinex-based methods [17], [22], which involve the intricate prior design, this network gets rid of the reliance on priors. The MSR-Net obtains the final enhanced image by fusing the results produced by the classical MSR model at multiple scales using an encoder–decoder network (refer to “Ours” in Fig. 1). As a result, the robustness of the MSR-Net is improved.

In addition, we design a hybrid convolution–axial attention block (HCAAB), which captures not only the local characteristics but also the global context through bringing together the convolution and self-attention mechanism [41], to address the limitations of the existing CNN-based methods [27], [28], [29]. In particular, we design a lightweight axial self-attention mechanism for the sake of overcoming the quadratic complexity associated with the original self-attention mechanism. Both the CC-Net and MSR-Net are built based on a set of HCAABs.

To our knowledge, either the CC-Net or the MSR-Net has not been explored for the UIE task before. To summarize, the contributions of this research can be identified as threefold.

- 1) We explicitly divide the complicated UIE task into color correction and visibility enhancement procedures and propose a deep CCMSR-Net, which contains two subnetworks, i.e., the CC-Net and the MSR-Net, with regard to the two procedures. In contrast to their traditional counterparts, the CC-Net is more robust to various degraded images while the MSR-Net does not rely on a prior. Compared with other deep learning methods, we avoid directly learning the complicated nonlinear mapping from limited training data. As a result, our CCMSR-Net is able to adapt to diverse degraded underwater images.
- 2) We design an HCAAB, which combines convolutions with the lightweight axial self-attention mechanism that we introduce. This combination allows our CCMSR-Net, built on top of HCAABs, to capture both the local characteristics and the global context.
- 3) To evaluate the effectiveness of the proposed CCMSR-Net, we conduct extensive UIE experiments on five testing sets. The results provide the community with a series of benchmarks.

The remainder of this article is organized as follows. In Section II, we review the literature related to this research. We introduce the proposed CCMSR-Net in Section III. The experimental setup and results are reported in Section IV. We examine the application of the enhanced images produced by our method to different vision tasks in Section V. Finally, we draw our conclusion in Section VI.

II. RELATED WORK

A. Traditional Methods

In terms of whether or not the underwater image formation model (IFM) is used, the traditional UIE methods can be divided into restoration-based and enhancement-based methods.

1) *Restoration-Based Methods*: The restoration-based UIE methods are normally adopted on top of the IFM. These methods aim to estimate the physical parameters of the IFM, which describe the degradation of underwater images, for the purpose of restoring underlying clean images. Due to the similar degradation mechanisms of the images captured in the air under the foggy weather and the images acquired in the underwater environment, many methods were proposed based on the dark channel prior (DCP) [42], which was originally used for image dehazing.

To adjust the color distortion of underwater images, Chiang and Chen [43] used the wavelength compensation technique. Then the DCP was used to remove the haze in the images. Drews et al. [10] proposed the underwater DCP (UDCP) to restore the color of underwater images. The DCP was modified with respect to the characteristics of underwater scenes. In [44], the depth information was estimated based on the degree of blurriness and the DCP was adopted to estimate the background light.

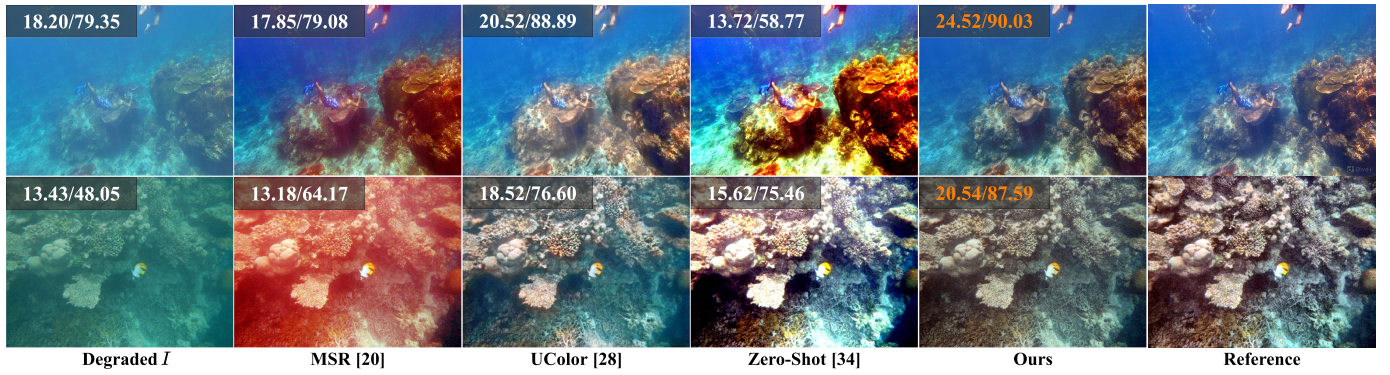


Fig. 1. Two sets of resultant images obtained from a real-world degraded underwater image in the UIEB [27] dataset. Within each set, a degraded image I , the enhanced images produced by MSR [20], UColor [28], zero-shot [34], and the proposed CCMSR-Net, and the reference image are displayed from left to right. The PSNR and SSIM [39] values are shown at the top-left corner of the degraded image or a resultant image, which are computed between this image and the reference image.

The generalized DCP (GDCP) was introduced by Peng et al. [11] through combining an adaptive color correction approach and the DCP for the sake of restoring the images with color cast and hazy degradation.

2) *Enhancement-Based Methods*: Basically, the enhancement-based methods are focused on adjusting the pixel values of underwater images order to enhance the visual quality without considering the IFM. For example, classic image enhancement methods, such as histogram equalization [45] and gamma correction [27], can be applied to some simple underwater scenes in which the degradation is nearly globally uniform. In [46], a multiscale fusion strategy is used to enhance the details based on the enhanced results in RGB color space and CIELab color space. Recently, Zhang et al. [15] proposed an enhancement method, namely, minimal color loss and locally adaptive contrast enhancement (MMLE), by adaptively balancing the color in the RGB color space and enhancing the contrast in the hue-saturation-intensity (HSI) color space. Wang et al. [47] performed UIE by aligning the color distributions of underwater images in the RGB and CIELab color spaces with those of natural images.

The Retinex theory, which was proposed to simulate the human perception of lightness and color, was also used to enhance the underwater images. Fu et al. [17] first transformed the color-corrected underwater images into the CIE-Lab color space and then performed the Retinex decomposition to derive the illumination and reflectance components. Each component was further enhanced using different strategies. In [16], an MSR model, which was designed on top of both the bilateral and trilateral filters, was used to enhance underwater images and suppress the artifact in the resultant images. Recently, Zhuang et al. [22] proposed a variational Retinex model, i.e., hyper-Laplacian reflectance priors (HLRP), which leveraged the hyper-Laplacian prior to enhance underwater images.

B. Deep-Learning-Based Methods

The deep-learning-based methods can be divided into supervised, semisupervised, and unsupervised methods according to different training strategies.

1) *Supervised Methods*: A large number of labeled data are normally required to train deep learning models using

supervised methods. Two different strategies can be used to construct a training dataset, including synthesizing fake underwater images using the terrestrial data and selecting the most visually pleasant image from a set of resultant underwater images enhanced using different UIE methods as the pseudo ground truth (i.e., the reference image). Li et al. [26] constructed a synthetic underwater image dataset using the indoor RGB images and the associated depth data. A UIE network was trained using this dataset.

In [27], a real-world dataset, namely, UIEB, was built using the second strategy. Inspired by the success of the fusion-based method [40], Li et al. [27] proposed the WaterNet, which was used to learn fusion weights for fusing different enhancement results. Jiang et al. [48] designed the LCNet which learned high-quality residuals and reduced the number of parameters by introducing Laplacian pyramids into the network. For the purpose of performing the UIE task in the coarse-to-fine manner, Cai et al. [29] proposed a cascaded deep UIE network.

2) *Semisupervised Methods*: To reduce the requirement of the labeled data, semisupervised learning techniques were also applied to the UIE task. Guo et al. [32] used the cycle-consistent generative adversarial network (CycleGAN) [30], which was originally proposed for unpaired image translation, to enhance underwater images. Li et al. [33] further used the structural similarity index (SSIM) loss function to improve the visual quality of the images enhanced using the CycleGAN. Fabbri et al. [31] first collected both severely degraded underwater images and relatively clear images. Then the CycleGAN was used to transfer the clear images to their degraded counterparts to construct a labeled training set. The underwater generative adversarial network (UGAN) was proposed, which was trained using the dataset, to enhance underwater images. Guo et al. [32] introduced the UWGAN, in which the multiscale dense block was designed to improve the generator.

In [49], an image translation framework and the contrastive learning technique were brought together for the purpose of deriving the more visual-pleasant results. Han et al. [1] proposed a lightweight encoder-decoder network for the UIE task and leveraged channel attention and spatial attention to learn the useful representations. Recently, Qi and Dong [50] proposed a physics-aware semisupervised UIE method which leveraged both the labeled and unlabeled data to train a

network. The parameters of the IFM, estimated using the network, were used to enhance degraded underwater images.

3) *Unsupervised Methods*: Unlike the supervised and semisupervised methods, which require the ground-truth data for training a network, unsupervised methods seek to exploit the general characteristics inherent in the unlabeled training data, to obtain their counterparts directly. Recently, Kar et al. [34] revealed the relationship between the input image and the corresponding controlled-perturbation image in the associated IFM parameters. In [34], a zero-shot image restoration approach was proposed for the purpose of restoring clear underwater scenes on top of this relationship and the multiple regularizations on saturation, color, and local consistency. In addition, Fu et al. [35] leveraged the relationship between the raw image and its redegraded counterpart for unsupervised underwater image restoration.

In addition, deep reinforcement learning techniques [51], [52] have been used for the UIE task, in which a reinforcement learning model was trained to leverage a set of image enhancement algorithms to produce the optimal result.

III. METHODOLOGY

Inspired by the achievements of the Retinex-based UIE methods, we decompose the UIE task into two stages, including color correction and visibility enhancement. Regarding the color correction stage, our emphasis is put on adjusting color distributions, to rectify color distortions. In terms of the visibility enhancement stage, we leverage the classical MSR model [20] to improve the visibility of color-corrected images.

To address those issues, we propose a novel deep CCMSR-Net. As shown in Fig. 2, this network contains a color correction subnetwork (CC-Net) and an MSR subnetwork (MSR-Net). For the purpose of performing color correction, the CC-Net uses a static weighting module (SWM) and a dynamic weighting module (DWM) to adjust color distributions. To further enhance the visibility, the MSR-Net is adopted based on the classic MSR model [20], which is used to estimate and eliminate the illumination. Since multiple results are produced by this model, a dynamic fusion module (DFM) is used to fuse them into the final enhanced image. Both the CC-Net and MSR-Net are built on top of an encoder-decoder network. As depicted in Fig. 3, this network is developed using a set of HCAABs that we design. In contrast to the existing CNN-based methods [27], [28], [29], these blocks capture both the local characteristics and the global context by jointly exploiting the convolution and self-attention mechanism [41].

A. Color Correction Network

According to the underwater IFM [7], the red channel suffers from the more severe attenuation than the green and blue channels. It has also been demonstrated that the blue channel will significantly degrade in the turbid water [53]. In contrast, the green channel retains the richer information. Therefore, it is important to compensate the red and blue channels for the purpose of correcting the distorted color. We are motivated to propose a color correction network (CC-Net), in order that both the channels can be adaptively compensated with the

green channel. Compared with the traditional color correction methods [14], [17], [40], which rely on predefined procedures and parameters, this network can be end-to-end trained and hence is more robust to diverse degraded images. As shown in Fig. 2, the CC-Net contains two parallel modules, i.e., an SWM and a DWM.

1) *Static Weighting Module*: Inspired by the traditional UIE methods [17], [40], the SWM is designed based on the differences between the red or blue channel and the green channel, to dominate the extent of information compensation. To be specific, this module generates two weight maps, i.e., W_R^S and W_B^S , by exploiting the rich information contained in the green channel. The two maps are used for the red and blue channels, respectively. Given a degraded image, the computation process of the SWM can be expressed as

$$W_R^S = (I_G - I_R) \odot (1 - I_R) \odot I_G \quad (1)$$

$$W_B^S = (I_G - I_B) \odot (1 - I_B) \odot I_G \quad (2)$$

where I_R , I_G , and I_B denote the red, green, and blue channels, respectively, and $\odot$ stands for the Hadamard product. The traditional UIE methods often follow a similar procedure based on the regionwise information extraction. Although this procedure can be advantageous, the region-based computation is less efficient and the region size chosen may affect the result. In this case, the adaptability of those methods is limited. To address this problem, we use a pixelwise approach in the SWM while aggregating the information contained in different regions using the DWM.

2) *Dynamic Weighting Module*: To increase the robustness of the color correction operation, we introduce the DWM, which implicitly learns a second set of weight maps. The DWM is appended to the SWM and is implemented as an encoder-decoder network (see Fig. 3). This module is able to capture the context information across different regions through the encoding process. Then the decoding process uses this information to allocate weights to individual pixels. This strategy enables the adaptable color correction operation across multiple regions. The output of the DWM is two additional weight maps: W_R^D and W_B^D .

3) *Color Correction*: The compensation operation for the red and blue channels is performed as

$$I'_R = I_R + W_R^S \odot W_R^D \quad (3)$$

$$I'_B = I_B + W_B^S \odot W_B^D. \quad (4)$$

Finally, I'_R , I_G , and I'_B comprise a color image, i.e., the color-corrected image I_{CC} . Due to both the SWM and DWM, the CC-Net is able to adaptively compensate the attenuated red and blue channels using the information contained in the green channel. The color-corrected image I_{CC} presents the more balanced color distribution than the degraded image I .

B. Multiscale Retinex Network

Although the CC-Net can reduce the distinct color distortion, other degradation phenomena still remain. To further increase the visibility, we introduce an MSR network (MSR-Net) on top of the MSR model [20]. This network aims to integrate the multiscale results produced by the MSR model

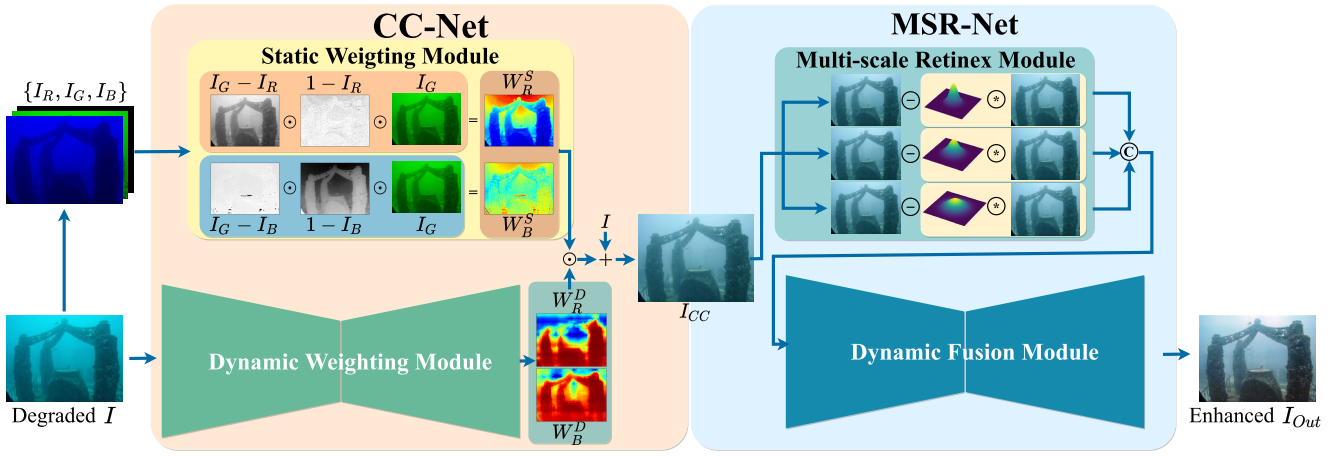


Fig. 2. Proposed CCMSR-Net, which comprises a color correction subnetwork (CC-Net) and an MSR subnetwork (MSR-Net). The CC-Net uses both the SWM and DWM to correct color distortion. Given that the corrected image is sent to the MSRM, the MSR-Net further enhances the visibility by dynamically fusing the output of this module.

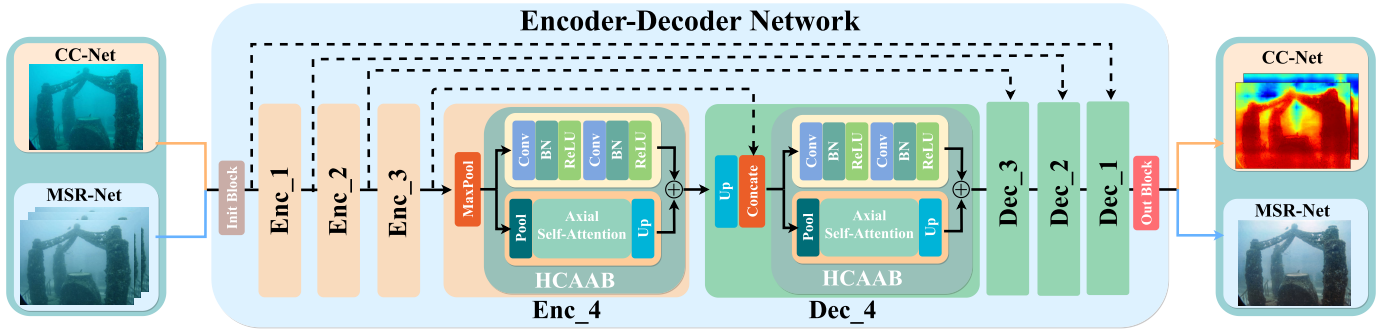


Fig. 3. Encoder-decoder network shared by both the DWM in the CC-Net and the DFM in the MSR-Net.

into an enhanced image. Despite some methods [17], [22] were adopted based on the Retinex model [18], they normally put their emphasis on formulating intricate priors for illumination estimation, while rarely consider the resilience of varied under-water images. In contrast, we use the efficient MSR model for illumination estimation and increase the robustness of our method through the utilization of deep learning techniques. In this section, we will first introduce the MSR model and then describe the MSR-Net in detail.

1) *Multiscale Retinex Model*: In terms of the Retinex [18] model, an image I can be treated as the Hadamard product of the reflectance R and the illumination L

$$I = R \odot L. \quad (5)$$

The reflectance R represents the underlying clear image while the illumination L is the factor which causes the decrease in image quality. To avoid the numerical error, the computation of illumination elimination is usually conducted in logarithmic space [19]. According to the single-scale retinex (SSR) [19] model, the estimation of L is derived by applying a Gaussian filter to I . Subsequently, the reflectance R , i.e., the enhanced image, can be obtained by eliminating the illumination. The computation can be expressed as

$$R = \log(I) - \log(I * G) \quad (6)$$

where G is the Gaussian function and $*$ denotes the convolution operation.

In contrast, the MSR [20] model uses multiple SSR models to achieve the better performance. In this case, the reflectance R can be computed as

$$R = \sum_{n=1}^N w_n [\log(I) - \log(I * G_n)] \quad (7)$$

where N is the number of scales and w_n stands for the weighting factor. Therefore, the effect of the MSR model is associated with the choice of w_n . Some methods [17], [22] abandoned the usage of the Gaussian function. To estimate the illumination, a variational architecture [54] was adopted with fixed priors instead. However, the complicated computation hindered the practical application of those methods and the fixed priors reduced the robustness to diverse degraded images.

2) *Multiscale Retinex Network*: We are inspired to introduce an MSR network (MSR-Net) for the purpose of increasing the visibility by estimating and eliminating the illumination. This network can be end-to-end trained from degraded images, rather than relying on designing sophisticated priors [17], [22]. As displayed in Fig. 2, the MSR-Net contains two serial modules, i.e., the MSR module (MSRM) and the DFM.

The MSRM leverages the basic principle of the MSR, described in (7), to generate multiple R maps, which are used as the source of the enhanced result. Thus, this module does not require complicated priors. However, the fusion process

demonstrated in (7) is implemented using the DFM. The input of this module is the concatenation of the R maps. The output is the final enhancement image, denoted as I_{out} . Through the proposed DFM, we fulfill the adaptive fusion of the R maps and derive the final enhancement result. It should be noted that the DFM uses a different set of weights from those used by the DWM in the CC-Net even though they share the same architecture.

C. Encoder–Decoder Network

The encoder–decoder network is used by both the DWM in the CC-Net and the DFM in the MSR-Net. This network is built using the proposed HCAAB.

1) *Encoder*: As shown in Fig. 3, an *Init Block* is first used to transform the input into a set of initial feature maps at the same resolution as that of the input. In total, the network uses four encoder blocks, denoted as Enc_1, Enc_2, Enc_3, and Enc_4, to progressively map the initial feature maps into the latent space.

2) *Decoder*: Four decoder blocks, denoted as Dec_4, Dec_3, Dec_2, and Dec_1, are symmetrically built. These blocks gradually upsample and decode the latent representation. In addition, skip connection is used to pass the multiscale feature maps from an encoder block to the corresponding decoder block, which is useful for deriving the finer grained prediction. An *Out Block* is appended to the last block of the network, i.e., Dec_1. This block is used to transform the output of Dec_1 to the desired prediction. In terms of the DWM in the CC-Net and the DFM in the MSR-Net, the *Out Block* is implemented as the combination of a convolutional layer and the Sigmoid activation function and a convolutional layer, respectively.

D. Hybrid Convolution–Axial Attention Block

The existing UIE methods [27], [28], [29] were normally developed based on CNN blocks. However, it is known that CNNs use limited effective receptive fields [36]. It has been illustrated that a large scope of context is crucial to the UIE task [55]. Recently, the transformer-based methods [41] achieved the competitive performance in many vision tasks [56], [57], [58]. The key component of transformers is the self-attention mechanism, which enables the network to exploit a global receptive field. As a result, the transformer network is able to capture long-range dependencies. Motivated by these studies, we propose a novel HCAAB, which brings the CNN and the transformer together, to address the limitations of the CNN-based methods.

1) *Hybrid Convolution–Axial Attention Block*: As depicted in Fig. 3, the HCAAB consists of two parallel subblocks, i.e., a CNN subblock and a transformer subblock. The CNN subblock comprises two consecutive 3×3 convolutional layers. Each layer is followed by a batch normalization (BN) layer and a rectified linear unit (ReLU) activation function. The objective of the CNN subblock is to extract the local characteristics from the image. In contrast, the transformer subblock facilitates the utilization of the global context information by exploiting the lightweight axial self-attention mechanism that we deliberately

design. Finally, the outputs of both the subblocks are fused through an elementwise addition operation.

2) *Lightweight Axial Self-Attention Mechanism*: Given an input $X \in \mathbb{R}^{H \times W \times C}$, the self-attention mechanism of vanilla transformer [41] is computed as

$$(Q_i, K_i, V_i) = (X_i W^Q, X_i W^K, X_i W^V) \quad (8)$$

$$Y_i = \text{Attention}(Q_i, K_i, V_i) \\ = \text{SoftMax}\left(\frac{Q_i(K_i)^T}{\sqrt{d}} + B\right) V_i \quad (9)$$

where Q , K , and V represent the query, key, and value feature maps, respectively, W^Q , W^K , and W^V stand for the learnable projection weights in terms of Q , K , and V , respectively, $i \in \{1, 2, \dots, HW\}$ indicates the location in the feature maps, B is the learnable position embedding, and SoftMax is the softmax activation function. The diagram of the self-attention mechanism is also shown in Fig. 4(a). It can be observed that the self-attention mechanism has the quadratic complexity with regard to the resolution of the input. In this case, the computational cost will be heavy when the resolution of the input is high.

Considering the recent efforts made on improving the efficiency of Transformers [56], [59], we are inspired to introduce a lightweight axial self-attention mechanism, for purposes of reducing the demand on both the computational and memory resources. Fig. 4(b) presents the diagram of the lightweight axial self-attention mechanism. To be specific, the input is first downsampled to a relatively small resolution, for example, 16×16 pixels. Then the lightweight axial self-attention is performed along the horizontal and vertical directions instead of operating on all the pixel positions. To be specific, we apply the average pooling operation along the horizontal and vertical directions separately, after the query, key, and value feature maps have been derived according to (8). The results are two sets of query, key, and value feature maps, denoted as $\{Q_i^h, K_i^h, V_i^h\}$ and $\{Q_i^v, K_i^v, V_i^v\}$ ($i \in \{1, 2, \dots, W\}$, $j \in \{1, 2, \dots, H\}$), respectively.

Furthermore, the attention mechanism can be conducted as

$$Y_i = \text{SoftMax}\left(\frac{\widetilde{Q}_i^h \widetilde{K}_i^h}{\sqrt{d}}\right) V_i^h + \text{SoftMax}\left(\frac{\widetilde{Q}_i^v \widetilde{K}_i^v}{\sqrt{d}}\right) V_i^v \quad (10)$$

where $\widetilde{Q}_i^h$, $\widetilde{K}_i^h$, $\widetilde{Q}_i^v$, and $\widetilde{K}_i^v$ are the horizontal and vertical query and key feature maps with the axial position embedding [59] and d denotes the number of channels of the feature maps. Compared with the self-attention mechanism [41], the proposed lightweight axial self-attention mechanism significantly reduces the computational and memory complexity by decomposing the self-attention computation into the horizontal and vertical attentions while effectively capturing the global context. Finally, the resultant feature maps are upsampled using the bilinear interpolation operation to the same resolution as that of the output of the CNN subblock.

IV. EXPERIMENTS AND RESULTS

To validate the effectiveness of the proposed CCMSR-Net, we conducted a series of experiments. In this section, we will

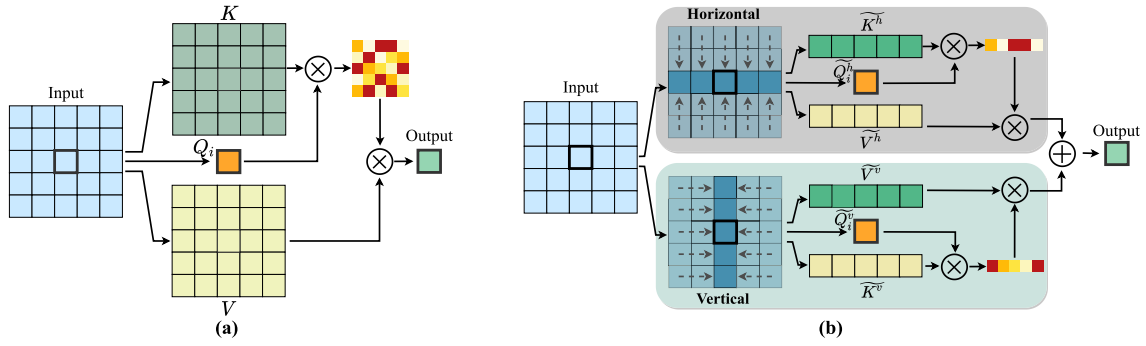


Fig. 4. (a) Self-attention mechanism [41] and (b) proposed lightweight axial self-attention mechanism. The former conducts the computation across all the pixel positions, which results in high memory and computational demands. In contrast, the latter squeezes the number of features and restricts the computation only in both the horizontal and vertical directions, which greatly reduces the memory and computational demands.

report the experimental setup and results in detail. In addition, we carried out an extensive ablation study, which will be introduced in this section.

A. Experimental Setup

We will first introduce the implementation details of the proposed method. Then we describe the datasets, baselines, and performance measures used in our experiments.

1) *Implementation Details*: We used PyTorch to implement our method. All the experiments were conducted on a GeForce RTX 3080 graphics processing unit (GPU). We trained the proposed CCMSR-Net using the AdamW [60] optimizer. The learning rate was set to 0.0001. Training images were resized to the resolution of 256×256 pixels before they were fed into the network. The mini-batch size was set to 12. We trained the CCMSR-Net for 300 epochs. The mse loss function was used. Both the color-corrected image I_{CC} and the final enhanced image I_{Out} were supervised by the ground-truth (reference) image I_{GT} . In this case, a combined loss function was built as

$$L = \lambda \cdot \text{mse}(I_{CC}, I_{GT}) + \text{mse}(I_{Out}, I_{GT}) \quad (11)$$

where the weighting factor λ was set to 0.8.

2) *Datasets*: The experiments were carried out on three real-world UIE datasets, including *UIEB* [27], *RUIE* [8] and *SUIM-E* [9], and a synthetic dataset, i.e., *EUVP* [61].

With regard to the real-world datasets, we followed the practice that Qi and Dong [50] performed. To be specific, the utilization of these datasets was described as follows. The *UIEB* [27] dataset was split into the training, validation, and testing sets, which contained 720, 80, and 80 images, respectively. The testing set was denoted as Test-U80. In addition, 60 more challenging images, which did not have reference images, contained in the *UIEB* dataset, comprised a second testing set, denoted as Test-C60. For the *RUIE* [8] dataset, the underwater color cast subset (UCCS), which contained 300 images, was used as a third testing set, namely, Test-UCCS. The official testing set of the *SUIM-E* [9] dataset was used as the fourth testing set, which contained 110 images. This testing set was referred to as Test-S110.

In terms of the synthetic dataset, we used 2185 images contained in the Underwater Scenes subset of the *EUVP* [61] dataset. Among these images, 2000 images were used for training and the rest were used for testing. This testing set was denoted as Test-Scenes.

3) *Baselines*: We compared the proposed method with 11 existing UIE methods, including two restoration-based methods, four enhancement-based methods, three supervised deep learning methods, and two unsupervised learning methods. Regarding the restoration-based methods, the UDCP [10] and GDCP [11] were used. The enhancement-based methods included the MSR [20] method, the Underwater-Retinex [17] approach, the MMLE [15] method, and the HLRP [22] method. For the supervised deep learning methods, we used WaterNet [27], UColor [28], and a recently proposed physics-aware method, namely, PA-UIENet [50]. Besides, the unsupervised learning methods consisted of the zero-shot [34] and USUIR [35] approaches.

4) *Performance Measures*: To evaluate the performance of different methods, we used three full-reference metrics, including peak signal-to-noise ratio (PSNR), SSIM [39] and learned perceptual image patch similarity (LPIPS) [62], and two nonreference metrics, i.e., underwater image quality measure (UIQM) [63] and underwater color image quality evaluation (UCIQE) [64]. In terms of these metrics except LPIPS, the higher value indicates the better visual quality. Besides, the number of parameters, floating-point operations (FLOPs), and inference latency were calculated as computational performance measures, given that a GeForce RTX 3080 GPU was used.

B. Experimental Results

In this section, we report the results derived in terms of the full-reference and nonreference metrics, respectively. A qualitative analysis is also provided.

1) *Full-Reference Metrics*: We compared the proposed CCMSR-Net with 11 baselines on three testing sets, including Test-U80, Test-S110, and Test-Scenes. Since these testing sets contained the ground-truth data, three full-reference metrics were used to measure the performance of these methods. The results are reported in Table I. It can be seen that our CCMSR-Net achieved the best performance with regard to the PSNR, SSIM, and LPIPS metrics across all the three testing sets. Although the WaterNet [27] fused the results of multiple UIE methods while the UColor [28] incorporated the depth information estimated using GDCP [11], they were still inferior to the proposed method. Compared with the traditional Retinex-based methods such as Water-Retinex [17], MSR [20],

TABLE I

COMPARISON OF DIFFERENT METHODS ON THE TEST-U80, TEST-S110, AND TEST-SCENES TESTING SETS IN TERMS OF THREE FULL-REFERENCE METRICS AND TWO NONREFERENCE METRICS. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN **RED BOLD** AND *Blue Italic* FONTS, RESPECTIVELY. THIS CONTINUES IN THE FOLLOWING TABLES

Method	Test-U80					Test-S110					Test-Scenes				
	PSNR↑	SSIM↑	LPIPS↓	UIQM↑	UCIQE↑	PSNR↑	SSIM↑	LPIPS↓	UIQM↑	UCIQE↑	PSNR↑	SSIM↑	LPIPS↓	UIQM↑	UCIQE↑
UDCP [10]	9.37	33.62	0.51	2.69	<i>0.0079</i>	10.04	34.28	0.48	1.23	0.0047	13.00	39.71	0.44	1.68	0.0031
GDCP [11]	12.46	67.41	0.46	2.17	0.0085	14.67	71.29	0.37	2.12	<i>0.0059</i>	13.90	63.31	0.42	1.89	<i>0.0035</i>
MSR [20]	16.60	76.66	0.33	2.97	0.0076	14.11	73.98	0.32	1.91	0.0056	12.27	65.34	0.45	<i>3.92</i>	0.0031
Water-Retinex [17]	18.12	77.89	0.31	4.55	0.0069	16.14	74.40	0.35	3.41	0.0053	15.68	68.38	0.44	4.33	0.0033
MMLE [15]	18.46	76.42	0.32	<i>3.98</i>	0.0070	17.23	76.88	0.29	2.65	0.0055	14.60	61.03	0.32	3.44	0.0033
HLRP [22]	13.36	21.80	0.44	3.48	0.0077	12.52	29.20	0.45	<i>2.89</i>	<i>0.0059</i>	11.91	17.89	0.50	3.86	0.0034
WaterNet [27]	17.02	70.27	0.54	2.11	0.0067	18.80	75.19	0.46	1.47	0.0050	25.50	83.93	0.23	3.69	0.0028
UColor [28]	20.93	85.19	0.25	3.44	0.0070	20.23	84.23	<i>0.21</i>	2.30	0.0058	24.11	81.21	0.29	3.64	0.0027
PA-UIENet [50]	<i>22.82</i>	<i>88.87</i>	0.16	3.60	0.0072	<i>20.46</i>	<i>87.22</i>	0.13	2.35	0.0056	<i>25.79</i>	<i>85.96</i>	<i>0.17</i>	3.65	0.0028
Zero-Shot [34]	13.99	58.72	0.38	3.94	0.0078	13.42	53.62	0.37	1.44	0.0060	12.07	41.83	0.44	1.76	0.0037
USUIR [35]	16.72	73.56	<i>0.23</i>	3.53	0.0073	18.54	78.44	0.25	1.09	0.0054	19.26	74.42	0.34	2.39	0.0032
CCMSR-Net (Ours)	23.14	89.75	0.16	3.57	0.0076	22.43	89.60	0.13	2.33	0.0058	26.26	87.00	0.14	3.70	0.0027

TABLE II

COMPARISON OF DIFFERENT METHODS ON THE TEST-C60 AND TEST-UCCS TESTING SETS WITH REGARD TO TWO NONREFERENCE METRICS

Method	Test-C60		Test-UCCS	
	UIQM↑	UCIQE↑	UIQM↑	UCIQE↑
UDCP [10]	0.56	0.0106	0.03	0.0012
GDCP [11]	1.49	0.0093	1.24	0.0018
MSR [20]	1.88	<i>0.0108</i>	2.98	<i>0.0020</i>
Water-Retinex [17]	2.81	0.0091	4.35	<i>0.0020</i>
MMLE [15]	<i>2.76</i>	0.0102	<i>4.05</i>	0.0019
HLRP [22]	1.47	0.0108	3.89	0.0021
WaterNet [27]	1.28	0.0096	1.25	0.0013
UColor [28]	2.08	0.0102	3.22	0.0016
PA-UIENet [50]	2.02	0.0098	3.24	0.0016
Zero-Shot [34]	1.75	0.0109	2.99	0.0021
USUIR [35]	0.94	0.0094	0.99	0.0014
CCMSR-Net (Ours)	2.07	0.0105	3.53	0.0015

and HLRP [22], the CCMSR-Net outperformed them with large margins. In particular, the supervised deep learning methods, including WaterNet [27], UColor [28], and PA-UIENet [50], usually performed better than the unsupervised deep learning approaches, i.e., zero-shot [34] and USUIR [35].

2) *Nonreference Metrics*: Considering both the Test-C60 and Test-UCCS testing sets were provided without the ground-truth data, two nonreference metrics were used for assessing the performance of our method and the 11 baselines on these datasets. The model trained using the *UIEB* [27] dataset was used. The results are presented in Table II. In addition, the nonreference metrics were computed for the Test-U80, Test-S110, and Test-Scenes datasets. The results are reported in Table I. As can be seen, the performance of our method was slightly inferior to that of its best counterpart. However, this was also the case for other deep learning methods, such as WaterNet [27] and USUIR [35]. In the literature, it has been pointed out that both the UIQM [63] and UCIQE [64] metrics cannot exactly reflect the perception quality of enhanced images [27], [28] and may prefer the results of the traditional methods to those of deep learning methods [29]. These findings may explain the relatively low UIQM and UCIQE values produced by our method.

3) *Qualitative Analysis*: Given the five testing sets, the results derived using the 11 baselines and our method are

TABLE III

COMPARISON OF DIFFERENT DEEP UIE METHODS IN TERMS OF MODEL SIZE, COMPUTATIONAL COMPLEXITY, AND INFERENCE LATENCY

Method	#Params (M)	FLOPs (G)	Latency (s)
WaterNet [27]	1.09	71.42	0.0064
UColor [28]	148.04	1402.18	0.0155
PA-UIENet [50]	28.44	49.59	0.0116
Zero-Shot [34]	0.38	3.14	0.0052
USUIR [35]	0.23	14.81	0.0021
CCMSR-Net (Ours)	21.13	43.60	0.0297

shown in Figs. 5–9, respectively. As can be observed, the traditional methods normally did not produce satisfying results. For example, the MSR [20] method usually produced an image with the high brightness but the color was not realistic. Although the other traditional Retinex-based methods [17], [22] increased the contrast, the pale color was produced. The deep learning methods also encountered difficulties in generating promising results. The WaterNet [27] tended to introduce the additional color cast while the UColor [28] method performed poorly on the degraded images captured in the turbid water (see Fig. 9). The PA-UIENet [50] could not completely remove the color cast in degraded images. In contrast, our CCMSR-Net improved different degraded images and produced images with the natural and vivid color. Even if the UIQM [63] or UCIQE [64] value produced by our method was lower than that generated by some baselines, the images that our method enhanced still manifested the satisfying visual quality.

4) *Complexity Analysis*: As reported in Table III, our method has the moderate space complexity and time complexity. Regarding the inference latency, our method took around 0.0297 s to enhance a single image, which suggests a speed of 33 frames/s (FPS). It is also noteworthy that both the UColor [28] and WaterNet [27] use preceding steps to generate the supplementary information. In contrast, our method performs the UIE task in an end-to-end manner.

C. Ablation Study

To investigate the effect of different components of the CCMSR-Net, we conducted a series of ablation experiments. For simplicity, only the Test-U80 testing set was used.

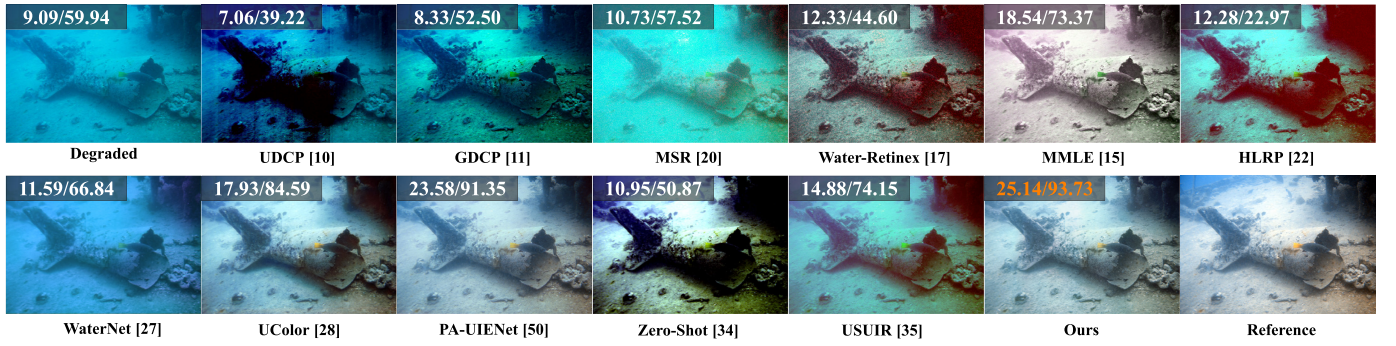


Fig. 5. Results produced by 11 baselines and our method in terms of one degraded images in the Test-U80 testing set. The PSNR and SSIM [39] values are shown at the top-left corner of the degraded image or an enhanced image, which are computed between this image and the reference image.

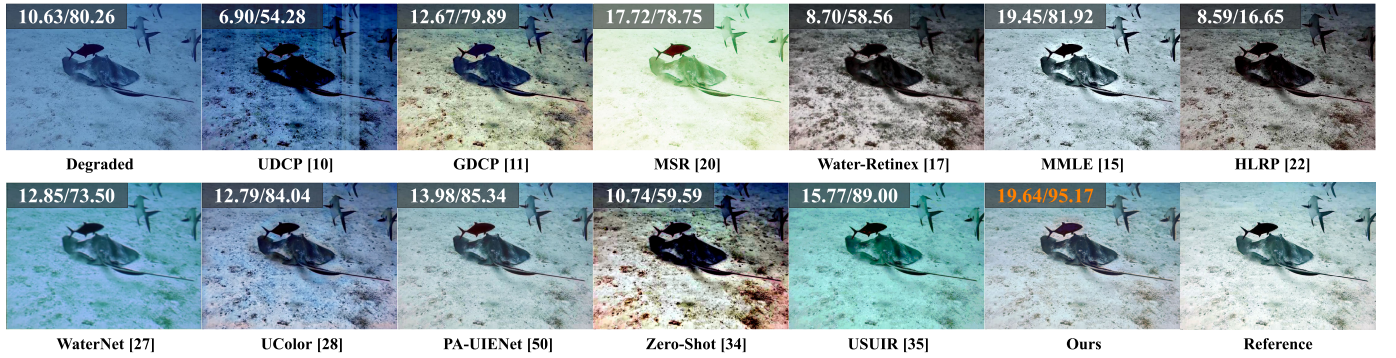


Fig. 6. Results produced by 11 baselines and our method in terms of one degraded images in the Test-S110 testing set. The PSNR and SSIM [39] values are shown at the top-left corner of the degraded image or an enhanced image, which are computed between this image and the reference image.

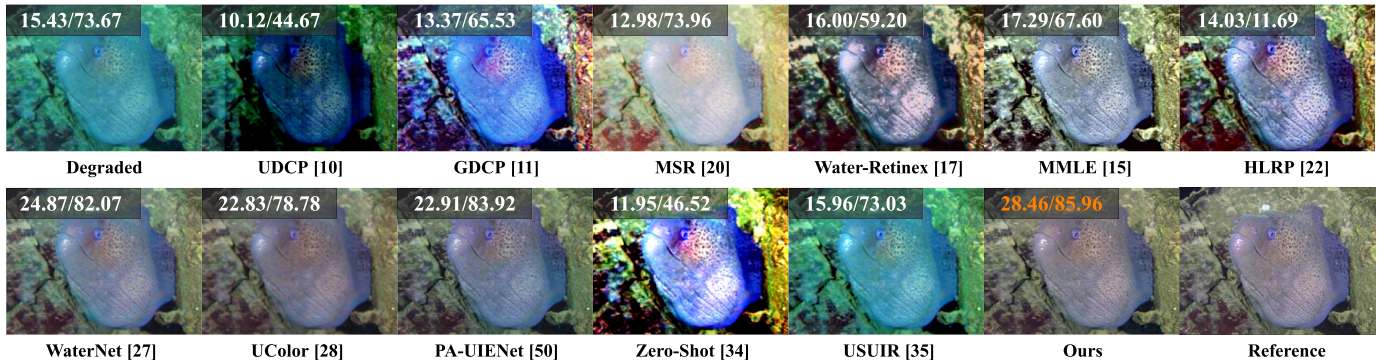


Fig. 7. Results produced by 11 baselines and our method in terms of one degraded images in the Test-Scenes testing set. The PSNR and SSIM [39] values are shown at the top-left corner of the degraded image or an enhanced image, which are computed between this image and the reference image.

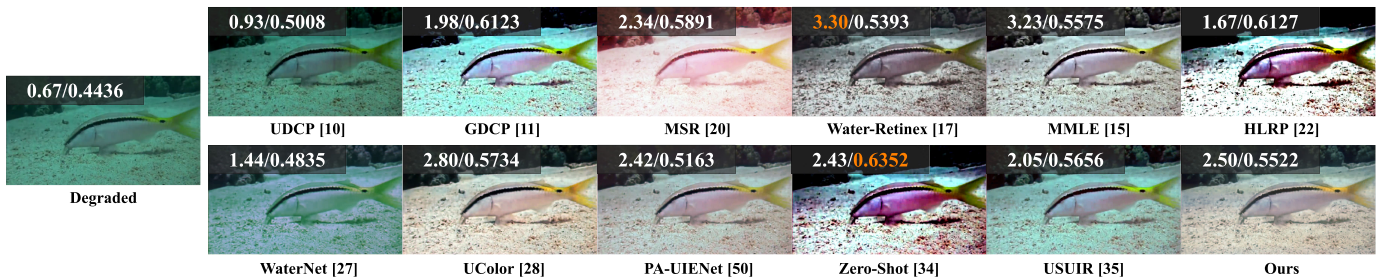


Fig. 8. Results produced by 11 baselines and our method in terms of one degraded images in the Test-C60 testing set. Both the UIQM [63] and UCIQE [64] values are shown at the top-left corner of the degraded image and enhanced images.

1) *Effect of the Subnetwork:* We investigated the effect of the subnetwork on the performance of the proposed CCMSR-Net by removing the MSR-Net or CC-Net separately. In this case, only the CC-Net or MSR-Net was remained. As reported in Table IV, either the CC-Net or MSR-Net produced the lower values than those generated by

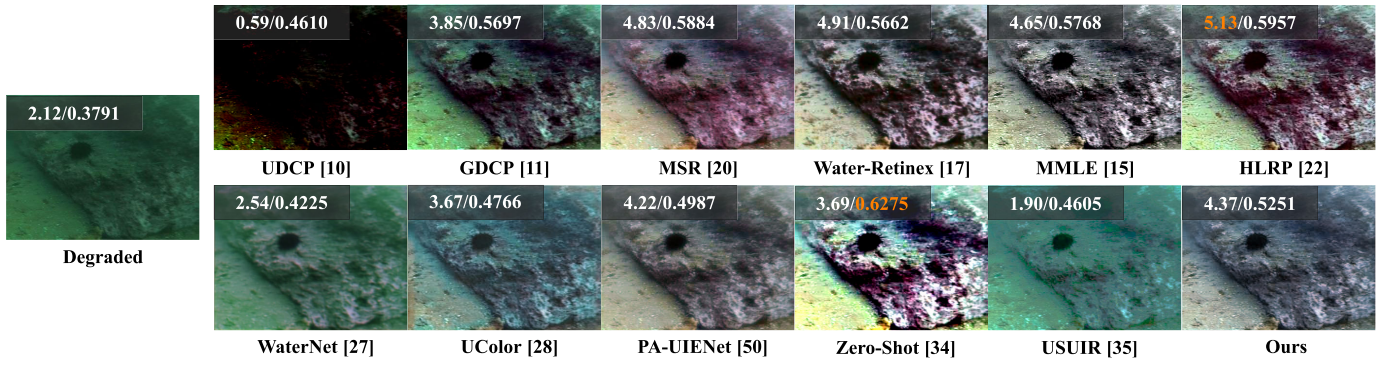


Fig. 9. Results produced by 11 baselines and our method in terms of one degraded images in the Test-UCCS testing set. Both the UIQM [63] and UCIQE [64] values are shown at the top-left corner of the degraded image and enhanced images.

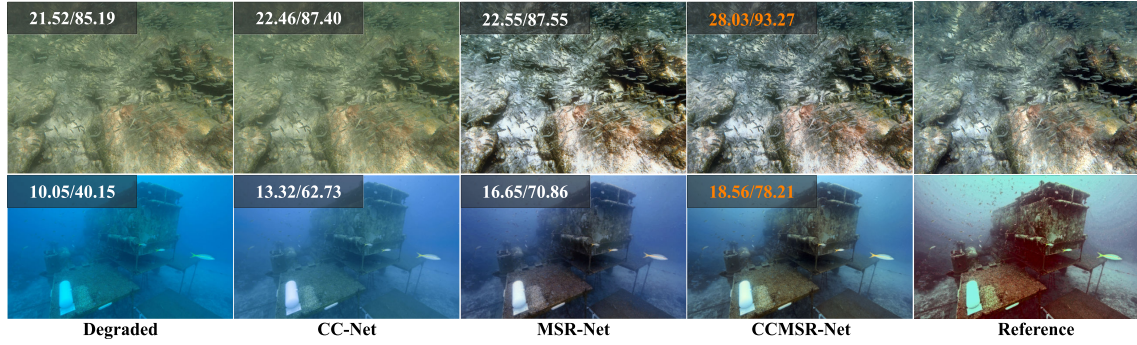


Fig. 10. Two sets of results obtained from two real-world degraded underwater images in the UIEB [27] dataset, respectively. Within each set, the degraded image, three enhanced images produced by the CC-Net, MSR-Net, and CCMSR-Net, and the associated reference image are displayed from left to right. The PSNR and SSIM [39] values, computed between a degraded or enhanced image and the reference image, are shown at the top-left corner of the image.

TABLE IV

COMPARISON BETWEEN EACH OF THE TWO SUBNETWORKS AND THE PROPOSED CCMSR-NET

	CC-Net	MSR-Net	CCMSR-Net (Ours)
PSNR $\uparrow$	17.23	22.81	23.14
SSIM $\uparrow$	78.65	87.88	89.75
LPIPS $\downarrow$	0.31	0.18	0.16
UIQM $\uparrow$	2.06	3.63	3.57
UCIQE $\uparrow$	0.0067	0.0075	0.0076

TABLE V

EFFECT OF THE COLOR CORRECTION METHOD TO THE PERFORMANCE OF THE CCMSR-NET

	w/o CC-Net	w/o dynamic weighting	w/ CC-Net (Ours)
PSNR $\uparrow$	22.81	22.64	23.14
SSIM $\uparrow$	87.88	89.34	89.75
LPIPS $\downarrow$	0.18	0.17	0.16
UIQM $\uparrow$	3.63	3.62	3.57
UCIQE $\uparrow$	0.0075	0.0074	0.0076

the CCMSR-Net (also see Fig. 10) which comprises both the subnetworks, in terms of the PSNR, SSIM, LPIPS, and UCIQE metrics. In particular, a significant performance drop was observed in the PSNR (23.14–17.23 dB), SSIM (89.75–78.65), and LPIPS (0.16–0.31) values when only the CC-Net was used. This finding should be attributed to the fact that the CC-Net was only designed for color correction while it could not remove the illumination.

2) *Effect of the Color Correction Method:* For the purpose of examining the effect of the color correction method on the performance of our CCMSR-Net, we removed the CC-Net or the DWM. In this case, we derived two variants of the CCMSR-Net, denoted as “w/o CC-Net” and “w/o dynamic weighting.” The results produced by the two variants and the CCMSR-Net are reported in Table V. As can be seen, the best performance was achieved by the CCMSR-Net together with the entire CC-Net with regard to the PSNR, SSIM, LPIPS, and UCIQE metrics. In particular, the PSNR, SSIM, and LPIPS values derived using our network without the CC-Net were

only 22.81 dB, 87.88, and 0.18, respectively. Given that the DWM was removed and only the SWM was used for color correction, the SSIM and LPIPS values reached to 89.34 and 0.17, respectively. The results were further improved when the DWM was added into the network.

To further assess the impact of the three color correction methods on color distribution equilibrium, we plot the histograms of the red, green, and blue channels of a degraded image and the enhanced images produced by the three methods in Fig. 11. As can be observed, the remaining network tended to retain the residual color distortion after the entire CC-Net had been removed. In the case that the SWM was used, the enhanced image showed the more balanced color distributions. However, this image suffered from the reduced color vibrancy due to the limited adaptability of the SWM. In contrast, our method achieved the comprehensive color distortion elimination and produced the vivid result with balanced color distributions, benefiting from the entire CC-Net.

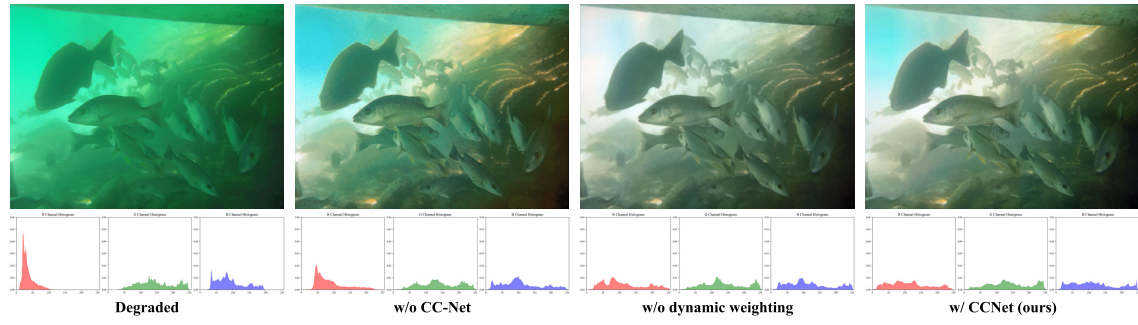


Fig. 11. Impact of color correction methods on balancing the distributions of different colors. Here, a degraded image contained in the UIEB [27] dataset and three associated enhanced images obtained using two variants of the CCMSR-Net and the CCMSR-Net itself are shown in the top row in turn. The histograms of the red, green, and blue channels of each image are displayed below this image.

TABLE VI

EFFECT OF THE BASIC BLOCK THAT THE CC-NET AND MSR-NET USE

CC-Net	MSR-Net	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	UIQM $\uparrow$	UCIQE $\uparrow$
CNN	CNN	21.90	88.34	0.19	3.56	0.0070
HCAAB	CNN	21.75	88.65	0.18	3.61	0.0070
CNN	HCAAB	22.57	89.26	0.18	3.64	0.0073
HCAAB	HCAAB	23.14	89.75	0.16	3.57	0.0076

3) *Effect of the Basic Block*: To investigate the effect of the basic block on which the CCMSR-Net was built, the CNN subblock in the proposed HCAAB was used to build the CC-Net and/or the MSR-Net instead of the entire HCAAB. In total, we derived four variants of the CCMSR-Net based on different combinations of the blocks. As shown in Table VI, the three variants, which used the HCAAB in one or two subnetworks, performed better than that built only on top of the CNN subblock. In terms of the PSNR, SSIM, LPIPS, and UCIQE metrics, the best result was derived using our CCMSR-Net. It is noteworthy that, however, the variant which only used the CNN subblock still outperformed all the baselines except the PA-UIENet [50] (see Table I), regarding the PSNR, SSIM, and LPIPS metrics. This finding also implies the effectiveness of our progressive enhancement network.

V. APPLYING ENHANCED IMAGES TO VISION TASKS

To investigate the usefulness of the proposed method to downstream vision tasks, we applied the enhanced images produced by our method to four different tasks, including object detection, edge detection, key-point matching, and salient object detection. Correspondingly, the YOLOv5 [65], Canny [66], scale-invariant feature transform (SIFT) [67], and U²Net [68] algorithms were used for the four tasks, respectively. The results obtained in these tasks are shown in Fig. 12.

It can be seen from Fig. 12(a) that the performance of YOLOv5 [65] was boosted by applying the enhanced images in which the visibility of objects was increased. In particular, not only missing or wrongly identified objects were correctly detected but also the confidence of the objects detected was increased, compared with the results derived using degraded images. As shown in Fig. 12(b), our method significantly increased the performance of the edge detection task. This result should be due to the ability of our method to enhance

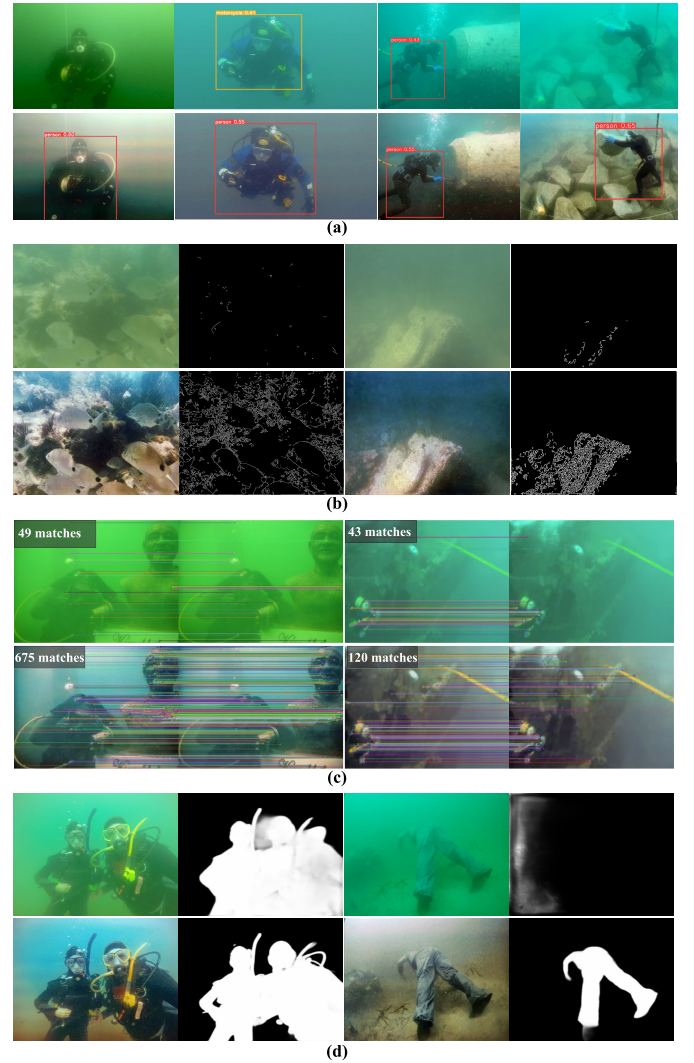


Fig. 12. Illustration of the usefulness of the enhanced images produced by our method for different vision tasks, including (a) object detection, (b) edge detection, (c) key-point matching, and (d) salient object detection, performed using YOLOv5 [65], Canny [66], SIFT [67], and U²Net [68], respectively. Within each group, the top row shows the results derived using degraded images while the bottom row displays the results obtained using the enhanced images.

the fuzzy details in degraded images. According to the results displayed in Fig. 12(c), the use of the enhanced images also improved the accuracy of key-point matching. As shown

in Fig. 12(d), U²Net [68] produced the more precise and fine-grained salient object detection results using the enhanced images produced by our method than those derived using degraded images. These results imply the usefulness of the proposed method, which is able to boost the performance of downstream vision tasks by enhancing the quality of degraded underwater images.

VI. CONCLUSION

In this article, we proposed a new deep CCMSR-Net. This network comprised a color correction subnetwork (CC-Net) and an MSR subnetwork (MSR-Net). The former was used to reduce the color distortion while the latter was designed for illumination estimation and elimination. Since the CC-Net could be end-to-end trained, it was more robust to diverse underwater images than the traditional color correction algorithms. On the other hand, the MSR-Net did not rely on the priors that the traditional Retinex-based methods used because it could also be end-to-end trained from degraded images. We designed an HCAAB, which combined convolutions and a lightweight axial self-attention mechanism. This block was used to build the CC-Net and the MSR-Net. In contrast to the CNN-based UIE methods, our CCMSR-Net was able to capture both the local characteristics and the global context. The CCMSR-Net performed better than, or at least comparably to, the 11 baselines tested in this study across five testing sets. We believe that this performance should be due to the effective combination of the color correction and illumination elimination operations, conducted using the CC-Net and the MSR-Net, respectively, which benefited from the joint exploitation of CNNs and transformers. Besides, the results produced by applying the enhanced images, which were derived using our method, to four vision tasks suggested that our method can be used as a preprocessing algorithm for downstream vision tasks, e.g., object detection and key-point matching.

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